

SOFTWARE REQUIREMENTS DOCUMENT (SRD)

VOXERA: Voice-Based Agentic AI Receptionist Platform

1. Project Overview

1.1 Project Title

VOXERA – Voice-Based Agentic AI Receptionist Platform

1.2 Project Description

Voxera is a voice-based Agentic AI platform that enables businesses to automate customer interactions through AI-powered receptionists. The platform can answer incoming calls, understand customer requests, provide business information, manage reservations, schedule appointments, send confirmations, and escalate conversations when required.

The system combines Speech-to-Text (STT), Speech Emotion Recognition (SER), Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and workflow automation to create human-like conversational experiences.

Businesses subscribe to Voxera and provide operational information such as services, pricing, FAQs, booking policies, and availability schedules. Voxera uses this information to serve customers 24×7 without requiring human intervention for routine tasks.

1.3 Objectives

The primary objectives of Voxera are:

- Automate customer call handling.
 - Reduce staffing costs.
 - Improve response availability.
 - Prevent missed bookings.
 - Provide 24×7 customer support.
 - Deliver personalized and emotion-aware conversations.
 - Enable seamless booking and scheduling workflows.
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2. Functional Requirements

FR-1 Incoming Call Handling

The system shall:

- Receive incoming phone calls.
- Connect calls through telephony providers.
- Support multiple concurrent callers.
- Record call metadata.

Outputs:

- Call ID
 - Caller information
 - Call start time
-

FR-2 User Voice Input

The system shall:

- Capture customer speech in real time.
- Support:
 - English
 - Hinglish
- Stream audio continuously.

Outputs:

- Audio stream
 - Session identifier
-

FR-3 Audio Preprocessing

The system shall perform:

- Noise reduction
- Silence trimming
- Audio normalization
- Sample rate standardization

Outputs:

- Cleaned audio stream

Metrics:

- Signal-to-noise ratio improvement
-

FR-4 Speech-to-Text Module

The system shall:

- Convert speech into text.
- Handle:
 - Accents
 - Code-switching
 - Pauses
 - Background noise

Outputs:

- Transcript
- Confidence score

Metrics:

- Word Error Rate (WER)

FR-5 Feature Extraction

The system shall extract acoustic features.

Traditional Features:

- MFCC
- Chroma
- Spectral Centroid
- Spectral Rolloff
- Zero Crossing Rate

Deep Features:

- Wav2Vec2 Embeddings
- HuBERT Embeddings

Outputs:

- Feature vector
-

FR-6 Speech Emotion Recognition

The system shall classify customer emotions.

Supported Emotions:

- Happy
- Sad
- Angry
- Neutral
- Fear
- Surprise

Outputs:

- Emotion label
- Confidence score

Example:

Emotion = Angry

Confidence = 87%

FR-7 Confidence Estimation

The system shall classify prediction confidence.

Ranges:

High Confidence:

- 80%

Medium Confidence:

- 50% – 80%

Low Confidence:

- < 50%

Outputs:

- Confidence category
-

FR-8 Emotion History Tracking

The system shall maintain emotional progression during calls.

Stored Data:

- Emotion labels
- Timestamps
- Confidence values

Example:

Angry → Neutral → Happy

FR-9 Conversation Context Memory

The system shall maintain conversation context.

Remember:

- Previous questions
- Previous responses
- Customer preferences
- Session information

Outputs:

- Context-aware responses
-

FR-10 Retrieval-Augmented Generation (RAG)

The system shall retrieve business-specific knowledge.

Supported Information:

- FAQs
- Pricing
- Room information
- Policies
- Service details
- Business rules

Pipeline:

Question

↓

Retrieve Documents

↓

Generate Response

Outputs:

- Relevant contextual information
-

FR-11 Emotion-Aware Response Generation

The system shall adapt responses based on detected emotions.

Inputs:

- Customer query
- Emotion state

Outputs:

- Personalized response

Example:

Angry Customer:

"Sorry for the inconvenience. Let me help resolve that."

Neutral Customer:

"How may I assist you today?"

FR-12 Tool Invocation Framework

The AI agent shall invoke external tools.

Supported Actions:

- Create booking
- Modify booking
- Cancel booking
- Send email
- Update spreadsheet
- Retrieve customer records
- Check availability

Outputs:

- Tool execution status

- Tool logs
-

FR-13 Reservation Management

The system shall:

- Create reservations
- Modify reservations
- Cancel reservations
- Verify availability

Outputs:

- Booking ID
 - Booking confirmation
-

FR-14 Calendar Integration

The system shall integrate with:

- Google Calendar
- Outlook Calendar

Capabilities:

- Slot booking
- Conflict detection
- Availability management

Outputs:

- Calendar event
-

FR-15 Email Notification System

The system shall:

- Send booking confirmations

- Send cancellation notices
- Send reminders

Outputs:

- Email status
 - Delivery confirmation
-

FR-16 Knowledge Base Management

Business administrators shall:

- Upload PDFs
- Upload FAQs
- Upload policies
- Update service information

Outputs:

- Searchable knowledge repository
-

FR-17 Text-to-Speech Module

The system shall:

- Convert generated responses into speech.
- Produce natural and human-like audio.

Outputs:

- Voice response

Requirements:

- Low latency
 - Natural pronunciation
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FR-18 Human Escalation

The system shall transfer calls to human staff when:

- Confidence is low
- User requests a human
- Critical issues arise

Outputs:

- Transfer status
 - Escalation logs
-

FR-19 Call Queue Management

The system shall:

- Manage multiple incoming calls
- Maintain waiting queues
- Provide wait-time estimates

Outputs:

- Queue status
-

FR-20 Commitment Acoustic Index (CAI)

The system shall calculate user engagement.

Inputs:

- Pitch variation
- Speaking rate
- Interruptions
- Pause duration
- Response length

Outputs:

CAI Score (0–100)

Interpretation:

0–30 → Low Engagement

30–70 → Moderate Engagement

70–100 → High Engagement

FR-21 Session Logging

The system shall store:

- Transcripts
- Emotions
- Confidence scores
- CAI scores
- Booking events
- Tool invocations
- Timestamps

Outputs:

- Session history
-

FR-22 Analytics Dashboard

The dashboard shall display:

Operational Metrics:

- Total calls
- Active calls
- Successful bookings
- Missed bookings

AI Metrics:

- Emotion distribution
- Average CAI
- Confidence scores

Business Metrics:

- Peak hours
 - Conversion rates
 - Average call duration
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FR-23 Multi-Tenant Business Management

The platform shall support:

- Multiple businesses
- Independent configurations
- Separate knowledge bases
- Separate booking systems

Outputs:

- Tenant isolation
-

FR-24 Subscription Management

The system shall support:

- Monthly subscriptions
- Usage tracking
- Plan upgrades
- Billing history

Outputs:

- Subscription status
-

FR-25 Voice Persona Configuration

Businesses shall configure:

- Male voice
- Female voice
- Formal tone
- Friendly tone

- Custom greetings

Example:

"Welcome to Hotel Paradise. How may I assist you today?"

FR-26 Dataset Management

Supported Datasets:

- MSP-Podcast
- MELD
- IEMOCAP
- RAVDESS

Capabilities:

- Import
 - Preprocessing
 - Training split
 - Validation split
-

FR-27 Model Training Module

The system shall support:

- Training
- Validation
- Testing

Metrics:

- Accuracy
 - Precision
 - Recall
 - F1 Score
-

FR-28 Explainability Module

The system shall explain emotion predictions.

Example:

Prediction: Angry

Contributing Factors:

- High pitch
 - Fast speaking rate
 - Increased vocal energy
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3. Non-Functional Requirements

NFR-1 Performance

- STT latency < 1 second
 - SER inference < 500 ms
 - Total response latency < 3 seconds
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NFR-2 Accuracy

- Emotion F1 Score $\geq 60\%$
 - WER $\leq 20\%$
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NFR-3 Scalability

Support:

- 10 concurrent calls (initial)
 - 100+ concurrent calls (future)
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NFR-4 Reliability

- System uptime $\geq 95\%$
 - Fault-tolerant operation
-

NFR-5 Availability

- 24×7 availability
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NFR-6 Security

Protect:

- Audio recordings
 - Transcripts
 - Business information
 - Customer records
-

NFR-7 Privacy

- Data anonymization
 - Consent-based storage
 - Secure deletion policies
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NFR-8 Maintainability

Modules shall be independently replaceable:

- STT
 - SER
 - RAG
 - TTS
 - Booking Engine
-

NFR-9 Portability

Support:

- Windows
 - Linux
-

NFR-10 Auditability

Maintain:

- Call logs
 - Booking logs
 - Tool execution logs
 - Escalation logs
-

4. System Architecture Requirements

Telephony Layer

- Incoming calls
 - Call routing
 - Queue management
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Speech Layer

- Audio capture
 - STT
 - TTS
-

Intelligence Layer

- SER

- RAG
 - LLM Agent
-

Tool Layer

- Booking engine
 - Calendar integration
 - Email automation
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Analytics Layer

- Dashboard
 - Reports
 - Monitoring
-

Storage Layer

- Session logs
 - Knowledge base
 - Business data
 - Booking records
-

5. Dataset Requirements

For each dataset specify:

- Source
- Size
- Labels
- Modality
- Usage

Example:

MSP-Podcast

Purpose:

Primary SER Training

Hours:

237+

Labels:

- Happy
- Sad
- Angry
- Neutral

Modality:

Audio

Usage:

Training and Evaluation

6. Evaluation Requirements

STT Evaluation

- WER
 - CER
-

SER Evaluation

- Accuracy
- Precision
- Recall
- F1 Score

RAG Evaluation

Human Evaluation:

- Relevance
- Helpfulness
- Empathy

Scale:

1–5

CAI Evaluation

Correlation Metrics:

- Pearson Correlation
- Spearman Correlation

Against:

- Customer Satisfaction
 - Engagement Score
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7. Future Scope

- Hindi support
- Punjabi support
- Regional language support
- Multilingual AI receptionist
- Video emotion recognition
- Facial emotion recognition
- CRM integration
- WhatsApp voice integration
- Mobile application
- Cloud-native deployment
- Multi-modal emotion recognition

- Healthcare receptionist agents
- Restaurant booking agents
- Government service helpline agents
- Enterprise call center deployment